SYSTEM BOOT: MEDBUDDY_OS_V1.0
STATUS: SECURE

MEDBUDDY: AI-Powered Medical Chatbot with Retrieval-Retrieval-Augmented Generation

Mini Project Report

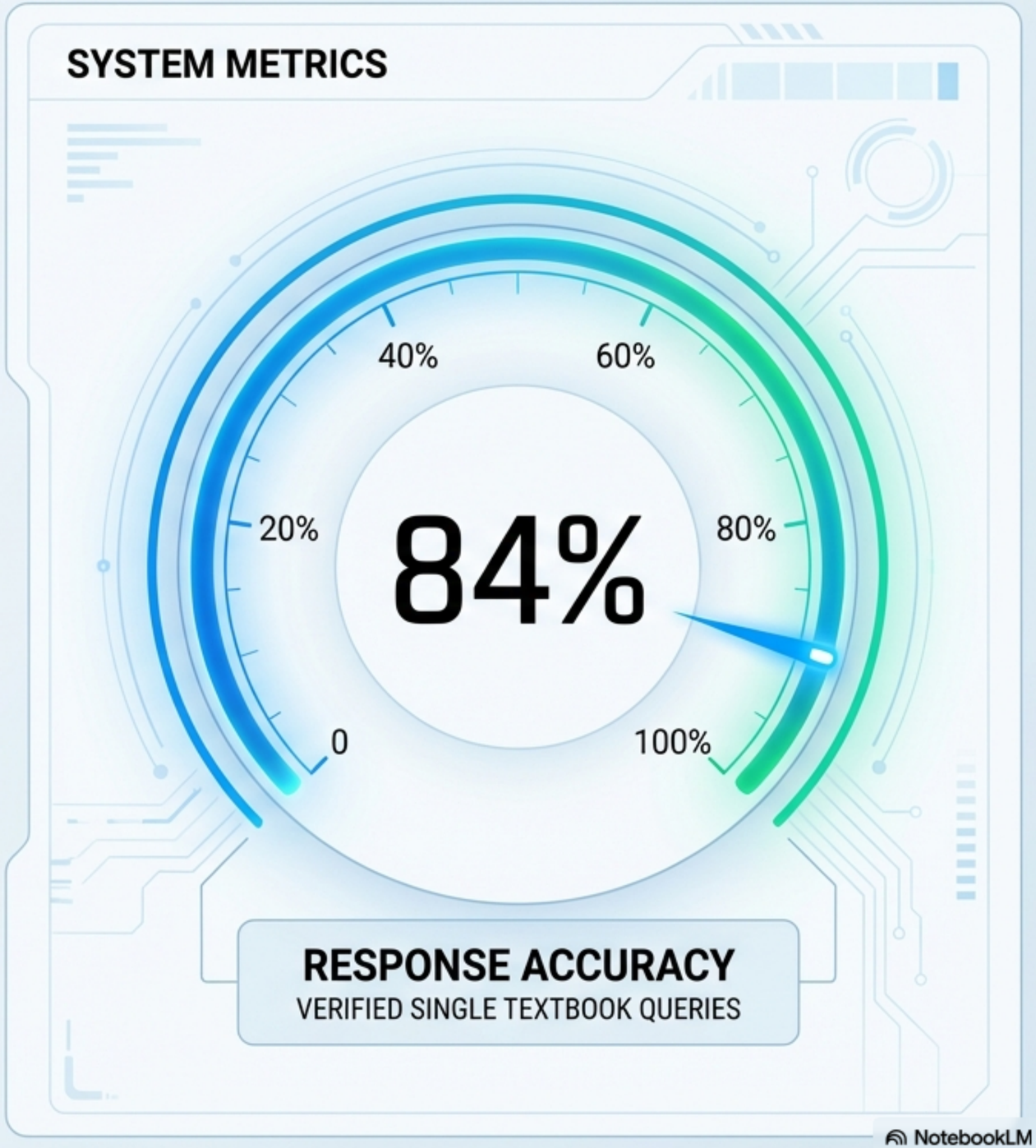
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MedBuddy: AI-Powered Medical Education Chatbot

- **MedBuddy** is an innovative **AI-powered medical chatbot** designed to revolutionize **medical education** and provide **evidence-based healthcare information** through advanced **Retrieval-Augmented Generation (RAG)** technology.
- By integrating **large language models** with **verified medical textbooks** and **research literature,** MedBuddy addresses critical challenges in **medical knowledge access, educational support,** and **diagnostic assistance.**
- The system leverages **Google Groq** as the foundational language model, augmented with **HuggingFace embeddings** for **semantic search,** and implements comprehensive **RAG pipelines** to ensure **response accuracy** of approximately **84%** for single textbook queries.



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The motivation behind Med-Buddy stems from the necessity to provide tangible assistance to medical students facing the **stress of massive curricula**. Existing systems often lack professional verification and reliable source-tracking.

To achieve this, the project is motivated by **three technical pillars**:

EMBEDDINGS



Turning human sentences into mathematical number-lists so the computer understands exact medical context.

VECTOR DATABASE



A specialized storage space for “number-words” allowing instant matching of user questions to verified textbook paragraphs.

LLM GROUNDING



A strict software rule forcing the AI to to only generate answers using the uploaded text, completely stopping outside fabrication.

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SI No.	Title [Year]	Author	Proposed Method	Disadvantages
1	AI Chatbots in Medical Education and Public Health [2023]	Baglivo et al., 2023; Eysenbach, 2023	AI tools effectively support vaccination training, enhance learning outcomes, and accelerate medical research cycles.	The emphasis remains on responsible AI integration with rigorous accuracy validation.
2	Retrieval-Augmented Generation (RAG) in Medical Chatbots [2024]	Bhavani Peri et al., 2024; Zhou et al., 2024	combine retrieval mechanisms with generative capabilities to ensure responses are grounded in authoritative medical sources.	RAG effectively eliminates hallucinations by constraining the generation space to retrieved, verified content.
3	Embedding Models and Semantic Search [2024]	Wang et al., 2024; Chen et al., 2024	These models employ self-knowledge distillation and multi-functionality approaches to achieve state-of-the-art semantic search performance.	The choice of embedding model directly impacts retrieval precision and, consequently, the quality of generated responses.
4	Large Language Models in Healthcare [N/A]	Studies demonstrate	LLMs excel at pattern recognition and knowledge synthesis	they require external grounding mechanisms to ensure medical accuracy.

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The existing system... operates as a **purely cloud-dependent** Retrieval-Augmented Generation (RAG) application. In this monolithic architecture, absolutely every stage of the NLP pipeline relies on the external API ecosystem.

While functionally capable, the existing cloud-only system suffers from several critical disadvantages that preclude its use in professional healthcare environments:

SEVERE DATA PRIVACY AND COMPLIANCE RISKS

Uploading proprietary medical documents, sensitive patient data, or internal hospital protocols to external, third-party servers is a massive security vulnerability.

EXORBITANT API COSTS

Generating embeddings for large volumes of data... consumes millions of tokens.



MONOLITHIC CLOUD STRUCTURE

HIGH LATENCY AND NETWORK DEPENDENCY

The system is entirely bottlenecked by internet bandwidth and the response times of the cloud provider.

RATE LIMITING

Cloud providers impose strict rate limits... Attempting to ingest a large medical corpus often results in API crashes and timeouts.

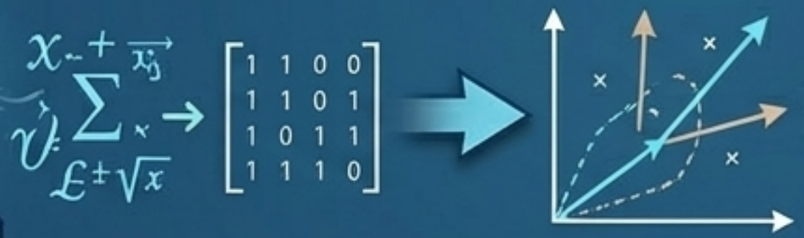
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The proposed system, **MedBuddy**, introduces a sophisticated, hybrid RAG framework that fundamentally restructures the processing pipeline to maximize both security and artificial intelligence capabilities. Recognizing the fatal flaws of the existing system, MedBuddy bifurcates the workload.

LOCAL / OFFLINE ZONE



By integrating mathematically efficient, lightweight open-source models—specifically variants like all-MiniLM-L6-v2—we enable MedBuddy to compute dense vector representations of complex medical jargon entirely offline.



This architectural contribution yields massive systemic improvements. First, it guarantees that no document data is ever sent to Google during the indexing phase, solving the primary privacy concern.

ONLINE / GROQ INFERENCE

Secure handoff for final text generation.



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[Placeholder: Base Paper Implementation]

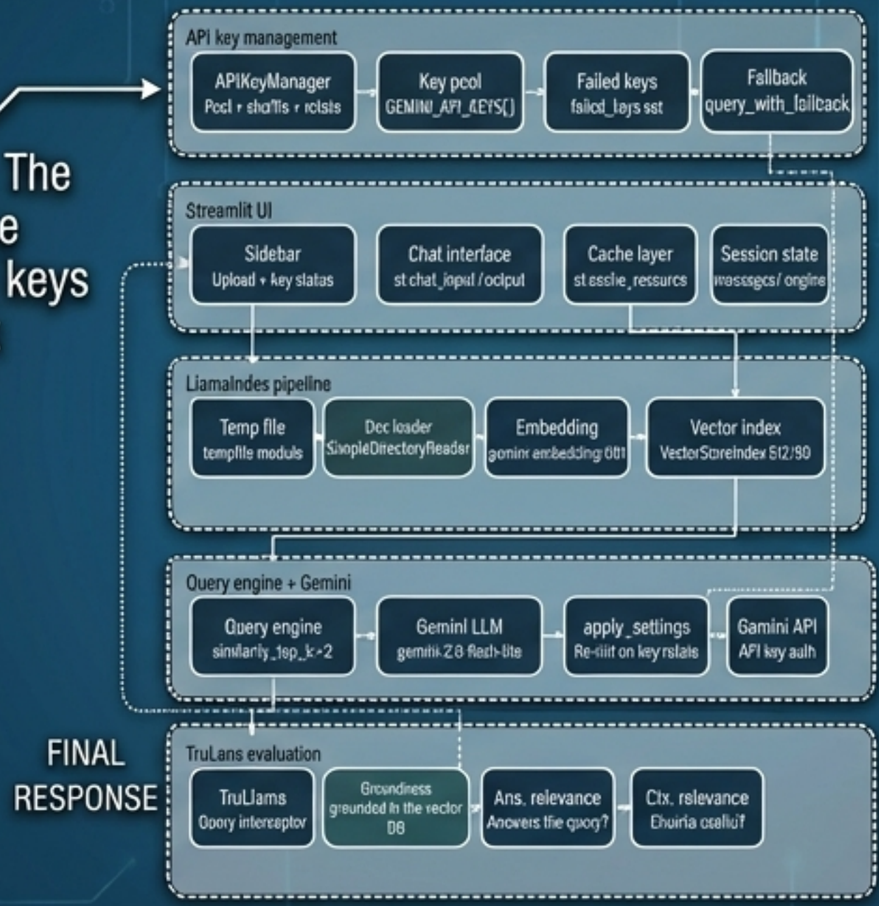
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[Placeholder: Our Contribution]

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The system architecture consists of 5 layers. The system architecture ensures that collected data transitions securely and meaningfully from raw PDF content to verified, context-grounded responses.

Layer 1 — API key management: The APIKeyManager class shuffles the key pool on startup, tracks failed keys in a set, and query_with_fallback automatically rotates and re-initialises whenever a quota error hits...



The updated implementation addresses these issues by offloading embedding computation to a local model while retaining a robust and high-performance language model through Groq, resulting in a more fault-tolerant and cost-efficient augmented architecture that balances local computation with external intelligence services.

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Streamlit
Powers the entire web-based interface, including the chat environment and session management.
LlamaIndex
Acts as the core data framework for document ingestion, text chunking (node parsing), and vector indexing.
Grok
Serves as a high-performance LLM provider to overcome the rate limits and API exhaustion found in the free tier.
FastEmbed
Provides local, efficient text embeddings, allowing the system to process data without relying on external cloud APIs.
TruLens
Performs real-time evaluation of AI groundedness and answer relevance to ensure a hallucination-free experience.

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Clinical Digital Interface – Results/Verification Dashboard

The initial design of Med-Buddy is build for proficiency and speed of data retrival, authenticity of the data and most importantly, for simple and convenient user-friendly experience through a conversational interface.



STATUS: VERIFIED

HYBRID RAG DEPLOYED

This improved implementation not only resolves previous bottlenecks but also establishes a foundation for future enhancements, confirming that the hybrid approach is highly successful.

The system emphasizes groundedness through RAG, effectively eliminating AI hallucinations and providing transparent, citation-backed responses.

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Clinical Digital Interface – Future Evolution & Scope

Ultimately, Med-Buddy aims to democratize safe access to complex medical knowledge, allowing students to study smarter and interview better by fundamentally shifting focus from data retrieval to true medical comprehension.

CURRENT PIPELINE
Hybrid Groq Dependency

100% AIR-GAPPED LOCAL ENVIRONMENT



The ultimate future scope for this project involves transitioning the architecture to a 100% offline, fully air-gapped system. This can be achieved by completely removing the Groq dependency and replacing it with local, open-source generative models such as Meta's Llama-3 or Mistral-7B. By utilizing advanced model quantization techniques... these heavy generative models can run locally.

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SYSTEM_INDEX // REFERENCES

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